Datasheet for the AMSMB HTR Dataset

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Abstract

This datasheet[1] describes a new dataset for the tasks of line detection and handwritten text recognition on medieval historical manuscripts. The dataset consists of 100 digitized documents written by 80 different hands and spanning three centuries (from 1208 to 1499). The manuscripts have been manually annotated by a medievalist paleographer. We are sharing the 100 images, their associated transcriptions in the PageXML format, and a csv metadata file.

1 Title

AMSMB HTR: A Dataset for Handwritten Text Recognition in Medieval Notarial Charters Written on Parchment (1208-1499).

2 Description

This is a new dataset for the tasks of line detection and handwritten text recognition on medieval manuscripts, focusing on notarial charters on parchment from the 13th to 15th centuries. The dataset is comprised of 100 digitized manuscripts (3,369 lines), carefully selected to represent the large variation that is present in the sources, encompassing at least 80 distinct hands and various document types (from sales and inventories to last wills and marriage contracts). Written primarily in Medieval Latin with fragments in Medieval Catalan, the manuscripts exhibit varying stages of preservation and degrees of deterioration. The dataset consists of 100 images and their associated transcriptions in the PageXML format, as well as a metadata csv file.

The Computational Archival History research line at the Computational Social Sciences and Humanities Laboratory (CSSH) of the Barcelona Supercomputing Center (BSC) focuses on creating datasets, training models, and developing tools for processing and extracting text from historical documents, making their content accessible to researchers and the general public. The current focus is on notarial documentation from the Late Middle Ages.

Repository The dataset is published on the Barcelona Supercomputing Center (BSC) Dataverse, available at: https://dataverse.bsc.es/citation?persistentId=perma:BSC/0VB0MC.

Papers and/or Other References The dataset and experiments are described in:

```
Mariona Coll Ardanuy, Iban Berganzo-Besga, Ramon Sarobe, and Coral Cuadrada. 2025. Evaluating Handwritten Text Recognition in Medieval Notarial Manuscripts: A New Dataset and Comprehensive Analysis. In International Conference on Document Analysis and Recognition.
```

Publisher Barcelona Supercomputing Center.

Point of contact Mariona Coll Ardanuy: contact at *mariona.coll@bsc.es*.

Supported Tasks This dataset has been created for the tasks of **line segmentation** (detecting lines of text in an image) and **handwritten text recognition** (transcribing the text in an image) on charters on parchment from the 13th to 15th centuries, written in Medieval Latin or Catalan in cursive script.

AI Category Computer Vision, Image-to-Text, Natural Language Processing.

Type of Cultural Heritage Application Transcribing; segmenting.

3 Distribution

Dataset Curators The following people were involved in creating the dataset:

[1] This datasheet is based on the second version of the 'Datasheet for Digital Cultural Heritage Datasets' template proposed in Alkemade et al. (2023).

- **Mariona Coll Ardanuy (BSC):** Lead of the *Natural Language Processing for the Humanities* research line at the Computational Social Sciences and Humanities Laboratory of the Barcelona Supercomputing Center. She is a computational linguist working in the application of natural language processing to humanities research. She conceptualized, designed, and supervised the creation and curation of the dataset, performed quality control and the formal analysis, and wrote the documentation.
- **Coral Cuadrada (BSC):** Lead of the *Computational Archival History* research line at the Computational Social Sciences and Humanities Laboratory of the Barcelona Supercomputing Center. Former professor of Medieval History, Historiography and Archival Science at the Universitat Rovira i Virgili, and director of the Archive of the Marquises of Santa Maria de Barberà (AMSMB). She coordinated the data collection and transfer, and oversaw the data selection process.
- **Ramon Sarobe (BSC):** Expert medievalist and paleographer working on the *Computational Archival History* research line at the Barcelona Supercomputing Center. He selected and transcribed the manuscripts, uploaded the transcriptions to eScriptorium, and wrote the transcription guidelines. He contributed the following metadata to the dataset: year of writing, name of the scrivener or notary, and document type based on content.

Licensing Information This dataset is published with a CC BY-SA 4.0 license: https://creativecommons.org/licenses/by-sa/4.0/.

Citation Information To cite this dataset:

```
Coll Ardanuy, Mariona; Cuadrada, Coral;
    Sarobe, Ramon, 2025, "Dataset for
    handwritten text recognition in
    medieval notarial charters written
    on parchment (1208-1499)", https://
    dataverse.bsc.es/citation?
    persistentId=perma:BSC/0VB0MC, BSC
    Dataverse, V1
```

Bibtex citation:

```
@data{0VB0MC_2025,
  author = {
    Coll Ardanuy, Mariona and
    Cuadrada, Coral and
    Sarobe, Ramon
  },
  publisher = {BSC Dataverse},
  title = {{Dataset for handwritten text
    recognition in medieval notarial
    charters written on parchment
    (1208-1499)}},
  year = {2025},
  version = {V1},
  doi = {BSC/0VB0MC},
  url = {https://dataverse.bsc.es/
    citation?persistentId=perma:BSC/0
    VB0MC}
}
```

Contributions The following people have also contributed to the creation and publication of this dataset: **Marc Prim** (archivist of the Arxiu Municipal de Vilassar de Dalt), who gave us access to the manuscripts; **Mercè Crosas**, director of the CSSH Laboratory at BSC, for project design and data strategy, **Joan Giner** (senior research engineer at BSC), who set up our local instance of the eScriptorium software and provided technical support; **Iban Berganzo-Besga** (senior research engineer at the BSC), who contributed in validating the dataset; **Felipe Gómez** (senior research engineer at BSC) and **Adrián Carrascosa** (research engineer at BSC), who ensured the dataset adherence to the FAIR principles by creating its instance on the BSC Dataverse.

4 Composition

The dataset consists of 100 images and their associated 100 PageXML files produced using the eScriptorium tool, as well as a metadata csv file.

Data Category Content and metadata.

Object Type Digitized medieval charters on parchment (images) and associated PageXML files; and a metadata csv file.

Dataset Structure The dataset is split into `train` (60 images, 60 xml files), `validation` (20 images, 20 xml files), and `test` (20 images, 20 xml files), and is accompanied by a `README.md` file and a `metadata.csv` file, with the following structure:

```
data/
|-- test/
|   |-- AMSMB_1-3-100.jpg
|   |-- AMSMB_1-3-100.xml
|   |-- ...
|   |-- AMSMB_C2-F3-72.jpg
|   |-- AMSMB_C2-F3-72.xml
|-- train/
|   |-- AMSMB_1-1-17.jpg
|   |-- AMSMB_1-1-17.xml
|   |-- ...
|   |-- AMSMB_C4-F14-37.jpg
|   |-- AMSMB_C4-F14-37.xml
|-- validation/
|   |-- AMSMB_1-62-39.jpg
```

```
|    |-- AMSMB_1-62-39.xml
|    |-- ...
|    |-- AMSMB_C3-F7-34.jpg
|    |-- AMSMB_C3-F7-34.xml
|-- metadata.csv
|-- README.md
```

The xml file uses the PageXML format,[2] generated using the eScriptorium[3] software (version 0.14.2), to describe the image page content. The xml expects to find the associated image in the same directory, as specified in the `imageFilename` attribute of the `Page` element in the xml, as in this example:

```
<Page imageFilename="AMSMB_1-1-17.jpg"
    imageWidth="3611"
    imageHeight="2838">
```

Data Fields The `metadata.csv` file associates each digitized image and corresponding PageXML file to the item metadata. It consists of a header and one row per image, with the following fields:

- **reference:** Reference identifier mapping the metadata to the digitized image and the transcription in the PageXML format. The reference is composed of the acronym of the archive and the signature of the manuscript (e.g., 'AMSMB_1-1-17').
- **split:** Either 'train', 'validation', or 'test'.
- **date:** Year when the manuscript was written.
- **archive:** Acronym of the archive.
- **signature:** Signature of the manuscript (e.g., '1-1-17').
- **scrivener/notary:** Name of the notary or scrivener who wrote the manuscript.
- **type:** Type of document (e.g., last will, oath, donation). Note that this field is in Catalan.
- **material:** Either parchment or paper (in this dataset, it is always 'parchment').
- **folder:** Physical folder where the physical manuscript is stored in the AMSMB archive.

Data Splits The dataset is split at the document-level to minimize the presence of lines written by the same hands in the different dataset splits. For more information about dataset splits, refer to Section 3.4 in Coll Ardanuy et al. (2025).

Languages The manuscripts are written primarily in Medieval Latin, but there is a strong linguistic influence from the vernacular Catalan language, both lexically (through loanwords and expressions) and syntactically (e.g., mirroring Catalan word order). In addition, Catalan is used instead of Latin in verbatim statements or utterances, and in specific documents such as wills and marriage contracts.

[2] https://github.com/PRImA-Research-Lab/PAGE-XML

[3] https://escriptorium.readthedocs.io/en/latest/

Table 1: Dataset summary: *images* is the number of digitized images; *lines* the overall number of lines; *13C*, *14C* and *15C* the number of lines from these centuries, and *hands* is the number of unique known hands.

	images	lines	13C	14C	15C	hands
train	60	2067	627	553	887	46
val	20	661	183	170	308	18
test	20	641	173	219	249	20
all	100	3369	983	942	1444	80

Descriptive Statistics The dataset consists of 100 digitized `jpg` images and 100 `xml` files in the `PageXML` format, as well as one `csv` file and a *readme* file. A summary of the dataset is provided in Table 1. The size of the dataset is 342 MB.

5 Data Collection Process

Curation Rationale Manuscripts were carefully sampled to represent, to the extent possible, the large variation that is present in the sources. They are written by at least 80 known hands (and several others unknown) across three centuries (from 1208 to 1499) throughout the Catalan-speaking lands. Their contents are classified into 29 different document types (e.g., last will, oath, donation). The documents are also diverse in terms of shape (with height-to-width ratios that range from 0.46 to 1.46), number of lines per charter (between 8 and 106 lines), and line lengths (with averages ranging from 84 to 341 characters). They display different stages of preservation, levels of brightness and contrast, and types of deterioration, such as holes, folds, stains, discoloured areas, and ink loss or fading.

Source Data The selected charters come from the Archive of the Marquises of Santa Maria de Barberà (AMSMB),[4] the most noteworthy private nobility archive in Catalonia. Its holdings include over 11K charters on parchment, of which 7.5K are from the 13th to the 15th centuries. In 1985, the archive began the pioneering process of describing its parchment collection using a bespoke software and database system that were commissioned to overcome the limitations of documenting the collections by hand. In the early 2000s, an agreement was signed with the town council of Vilassar de Dalt to make the archive's holdings available for

[4] https://archivemarquisesofbarbera.com

public consultation. This gave rise to an annual digitization campaign, starting with the parchment collection, the only one that had been fully described. Descriptions and images (in the TIFF format) were then integrated into a document management system to facilitate consultation to researchers. The archive is located at the castle of the Marquises of Santa Maria de Barberà in Vilassar de Dalt, home of the de Sarriera family, who financed all projects.

Preprocessing Given the large size of the documents, resulting in large TIFF files (images up to 650MB), we programmatically converted the TIFF files to JPEG files using the PIL library for python.

Annotations Documents have been transcribed by an expert paleographer with decades of experience working with medieval manuscripts. A locally installed instance of the eScriptorium software (version 0.14.2) has been used to manually align the transcription with the image. Lines have been automatically detected using the default *blla* segmentation model provided by Kraken[5] and eScriptorium, and have been manually reviewed and corrected.

Transcription Transcriptions follow a semi-diplomatic approach that expands abbreviations while respecting common transcription conventions for Medieval Latin and Catalan. The following points summarize our main decisions:

1. Allographs (variants of the same letter) are reduced to their underlying grapheme (the standardized representation). It is worth noting:
    - In Latin, the long 'i' (ressembling a 'j') is transcribed as 'i'. For example, we write '*vicarii*' instead of '*vicarij*'; and '*iurisdiccionis*' instead of '*jurisdiccionis*'.
    - In Latin, 'v' and 'u' are regularized according to convention. For example, we write '*reverendus*' and not '*reuerendus*'.
    - In Catalan, 'i' and 'j' are two distinct graphemes, and are transcribed as such. The same is true of 'u' and 'v'. For example, we write '*adjudicat*' and not '*adiudicat*'; and '*vint*' and not '*uint*'.
2. In general, grammatical or spelling irregularities are retained, and not corrected. However:
    - We use standard criteria for punctuation and capitalization for both Latin and Catalan: proper names and first words in a sentence are always capitalized. Besides, we follow standard criteria for diacritics and apostrophes in Catalan.
    - In Latin, the double 'f' at the beginning of a word is reduced to a single 'f'. For example, '*ffranciscus*' becomes '*Franciscus*' (capitalized, as already discussed).
    - The word 'Signum' (or any variation, such as 'Sig+num' or 'Sig++na', indicating the signatures of individuals) is always capitalized.
3. Proper names are not translated and are transcribed as they appear in the manuscript (but capitalized), respecting white spaces and spelling irregularities (except the initial double 'f', as mentioned), and respecting non-standard word separations. For example, we keep '*Fontisrubei*' instead of using the current toponym in Catalan (i.e., *'Font-Rubí'*).
4. Abbreviations and symbols are resolved to the full form of the word, with the exception of abbreviated first names, which are not resolved. For example, '*novint*' is resolved to '*noverint*', but *'B.'* is not resolved to the full name (typically '*Bernardus*' or '*Berengarius*').

When possible, our line segmentation respects the written line in the manuscript, even if there is a white space or illegible text in the line, except if there is a large gap or hole in the document (see Figure 1). The cross signature sign was also used to split the line into two different segments or lines (*'Sig'* on the one hand, and the rest of the line on the other), as shown in example in Figure 2.

Controlled Vocabularies We used a simplified version of the SegmOnto[6] vocabulary to describe the layout of the manuscripts. We annotated the following regions: (1) *MainZone*, (2) *MarginTextZone*, (3) *DropCapitalZone*, (4) *GraphicZone:Signum*, and (5) *GraphicZone:Other*. We distinguished between the following types of lines: (1) *DefaultLine*, and (2) *InterlinearLine*.

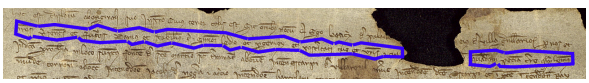


Figure 1: Segmentation (polygon in blue) of a line separated by a large gap in the manuscript, splitting the line in two parts. Reference: *AMSMB_C1-F2-10*.

[5]https://kraken.re/main/index.html

[6]https://segmonto.github.io/.

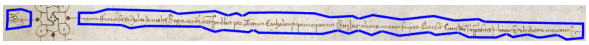


Figure 2: Segmentation (polygon in blue) of a line containing a *Sig+num* (cross signature) sign, splitting the line in two parts. Reference: *AMSMB_1-70-42*.

Maintenance The dataset will be maintained until 2028, including updates to the data.

6 Considerations for Using the Data

The dataset was created with the aim of training line segmentation and line-level transcription models for charters written on parchment in Gothic cursive script, in Medieval Latin or Catalan, from the 13th to 15th centuries. The dataset has not been tested for other uses, which may be out-of-scope.

Ethical Considerations The dataset does not include sensitive or personal data.

Discussion of Biases In the context of digital heritage collections, biases originate from the moment documents are created, collected, and selected for digitization (Beelen et al., 2023). Since this dataset was created for the task of handwritten text recognition, our sampling strategy focused on stratifying across characteristics such as image quality and shape, degree of preservation and types of deterioration, handwriting style, and the actual textual content. Several aspects need to be considered depending on the expected downstream use of this dataset. For example, a model trained on this dataset may be used to transcribe the rest of the parchment collection, which would enable longitudinal quantitative content-based analyses. Notarial collections are a unique source for the study of the past, because they allow us to access a significant part of the population that is often not present in other sources. However, it should be noted that charters written on parchments were usually the clean copies that were handed to the customers (unlike notarial protocols written on paper, which were the drafts kept by the notaries). Charters that have survived until today (and which are preserved in archives such as the AMSMB) have been gathered over the centuries, often through the incorporation of the holdings from aristocratic houses by means of inheritance and marriage, necessarily resulting in biases which need to be taken into account when using this data.

Potential Societal Impact This dataset is aimed at enabling the automatic transcription of medieval notarial charters on parchment. These documents record all kinds of social and economic activities, from marriage contracts and last wills to purchases and debts, allowing us to glimpse into the everyday lives of common people in the late Middle Ages, even documenting situations of inequality, discrimination or privilege. To a certain extent, notarial collections may enable a non-elitist perspective into life in the late Middle Ages. However, these documents still reflect the social and historical inequalities of the time (e.g., who had the means to go to the notary?, which documents were deemed important enough to preserve?), with the risk of reinforcing existing harmful biases if not handled properly.

Acknowledgments

The digitized images have been provided by the Arxiu Municipal de Vilassar de Dalt (AMVD), through its agreement with the Archive of the Marquises of Santa Maria de Barberà (AMSMB). Coral Cuadrada would like to thank His Excellency Mr. Ramon de Sarriera for facilitating public consultation of the AMSMB Parchment Fund. Mariona Coll Ardanuy acknowledges her AI4S fellowship within the "Generación D" initiative by Red.es, Ministerio para la Transformación Digital y de la Función Pública, for talent attraction (C005/24-ED CV1), funded by NextGenerationEU through PRTR.

References

Henk Alkemade, Steven Claeyssens, Giovanni Colavizza, Nuno Freire, Jörg Lehmann, Clemens Neudecker, Giulia Osti, and Daniel van Strien. 2023. Datasheets for digital cultural heritage datasets. *Journal of Open Humanities Data*.

Kaspar Beelen, Jon Lawrence, Daniel CS Wilson, and David Beavan. 2023. Bias and representativeness in digitized newspaper collections: Introducing the environmental scan. *Digital Scholarship in the Humanities*, 38(1):1–22.

Mariona Coll Ardanuy, Iban Berganzo-Besga, Ramon Sarobe, and Coral Cuadrada. 2025. Evaluating handwritten text recognition in medieval notarial manuscripts: a new dataset and comprehensive analysis. In *International Conference on Document Analysis and Recognition*.